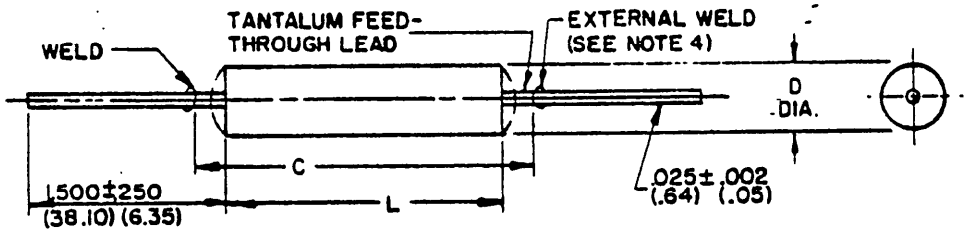
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (NONPOLARIZED, ETCHED FOIL) - 85°C (VOLTAGE DERATED TO 125°C), STYLES CL22 AND CL23

INACTIVE FOR NEW DESIGN; USE MIL-C-39006/2

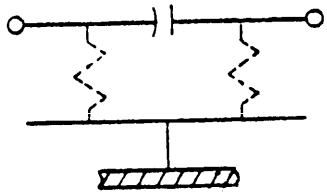
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



Case Size	Dimensions			
	L 1/ ±.062 (1.57)	D ±.016 (.41)	D Max	C Max
		CL22	CL23	
E1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)
G2	.969 (24.61)	.281 (7.14)	.312 (7.92)	1.469 (37.31)
G3	1.438 (36.53)	.375 (9.53)	.406 (10.31)	1.938 (49.23)
G4	2.125 (53.98)	.375 (9.53)	.406 (10.31)	2.625 (66.68)
G5	2.750 (69.85)	.375 (9.53)	.406 (10.31)	3.250 (82.55)

1/ For style CL23, nonshrinkable sleeving shall extend .016 inch minimum, .062 inch maximum, beyond each end of the case; shrinkable sleeving shall lap over the ends of the case.



CASE
CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Styles CL22 and CL23 capacitors.

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation 1/	Voltage			Capacitance	DC leakage (max)		Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C & 125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	ohms	
CL2-BE100UN E	15	10	17.2	10	2	4	339	E1
CL2-BE400UN G	15	10	17.2	40	2	10	85	G2
CL2-BE121UN G	15	10	17.2	120	3	40	27	G3
CL2-BE251UN G	15	10	17.2	250	6	80	14	G4
CL2-BE351UN G	15	10	17.2	350	9	100	10	G5
CL2-BG050UN E	25	15	28.8	5	2	4	680	E1
CL2-BG200UN G	25	15	28.8	20	2	13	170	G2
CL2-BG700UN G	25	15	28.8	70	3	40	49	G3
CL2-BG141UN G	25	15	28.8	140	6	80	24	G4
CL2-BG201UN G	25	15	28.8	200	10	100	17	G5
CL2-BH4R5UNE	30	20	34.5	4.5	2	6	754	E1
CL2-BH180UN G	30	20	34.5	18	2	14	188	G2
CL2-BH600UN G	30	20	34.5	60	4	50	57	G3
CL2-BH121UN G	30	20	34.5	120	8	90	28	G4
CL2-BH171UN G	30	20	34.5	170	11	120	20	G5
CL2-BJ2R5TN E	50	30	57.5	2.5	2	4	1355	E1
CL2-BJ100TN G	50	30	57.5	10	2	13	339	G2
CL2-BJ300TN G	50	30	57.5	30	3	40	113	G3
CL2-BJ600TN G	50	30	57.5	60	4	75	57	G4
CL2-BJ800TN G	50	30	57.5	80	5	100	41	G5
CL2-BL1R5TNE	75	50	86.2	1.5	2	4	2255	E1
CL2-BL060TN G	75	50	86.2	6	2	13	560	G2
CL2-BL150TN G	75	50	86.2	15	3	34	226	G3
CL2-BL350TN G	75	50	86.2	35	5	80	97	G4
CL2-BL500TN G	75	50	86.2	50	10	100	68	G5
CL2-BN010TN E	100	65	115.0	1	2	4	3385	E1
CL2-BN040TN G	100	65	115.0	4	2	13	845	G2
CL2-BN120TN G	100	65	115.0	12	3	37	283	G3
CL2-BN250TN G	100	65	115.0	25	5	75	135	G4
CL2-BN350TN G	100	65	115.0	35	7	100	97	G5
CL2-BQ0R5SN E	150	100	172.0	0.5	2	4	6780	E1
CL2-BQ020SN G	150	100	172.0	2	2	13	1693	G2
CL2-BQ060SN G	150	100	172.0	6	3	30	565	G3
CL2-BQ120SN G	150	100	172.0	12	5	80	283	G4
CL2-BQ180SN G	150	100	172.0	18	8	100	188	G5

1/ Complete type designation will include an additional symbol to indicate style.

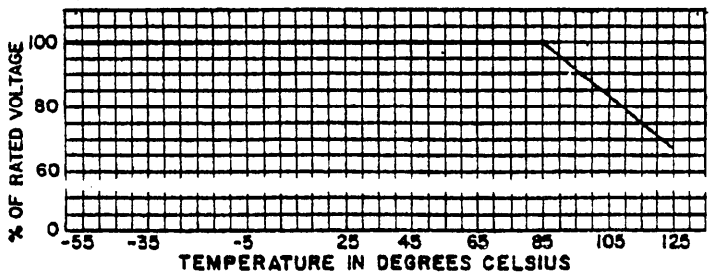


FIGURE 2. Voltage derating with temperature.

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TABLE II SUPERSESSION & CROSS REFERENCE DATA

SUPERSEDED		
FROM Type Designation 1/ MIL-C-3965/18D	TO Type Designation 1/ MIL-C-3965/18E	Cross Reference For New Design Part Number MIL-C-39006/2C
CL2-BE100UNE	CL2-BE100UNE	M39006/02-2232
CL2-BE400UNE	CL2-BE400UNG	M39006/02-1228
CL2-BE121UNE	CL2-BE121UNG	M39006/02-1229
CL2-BE251UNE	CL2-BE251UNG	M39006/02-1230
CL2-BE351UNE	CL2-BE351UNG	M39006/02-1231
CL2-BG050UNE	CL2-BG050UNE	M39006/02-2237
CL2-BG200UNE	CL2-BG200UNG	M39006/02-1232
CL2-BG700UNE	CL2-BG700UNG	M39006/02-1233
CL2-BG141UNE	CL2-BG141UNG	M39006/02-1234
CL2-BG201UNE	CL2-BG201UNG	M39006/02-1235
CL2-BH4R5UNE	CL2-BH4R5UNE	M39006/02-2242
CL2-BH180UNE	CL2-BH180UNG	M39006/02-1236
CL2-BH600UNE	CL2-BH600UNG	M39006/02-1237
CL2-BH121UNE	CL2-BH121UNG	M39006/02-1238
CL2-BH171UNE	CL2-BH171UNG	M39006/02-1239
CL2-BJ2R5TNE	CL2-BJ2R5TNE	M39006/02-2244
CL2-BJ100TNE	CL2-BJ100TNG	M39006/02-1240
CL2-BJ300TNE	CL2-BJ300TNG	M39006/02-1241
CL2-BJ600TNE	CL2-BJ600TNG	M39006/02-1242
CL2-BJ800TNE	CL2-BJ800TNG	M39006/02-1243
CL2-BL1R5TNE	CL2-BL1R5TNE	M39006/02-2249
CL2-BL060TNE	CL2-BL060TNG	M39006/02-1244
CL2-BL150TNE	CL2-BL150TNG	M39006/02-1245
CL2-BL350TNE	CL2-BL350TNG	M39006/02-1246
CL2-BL500TNE	CL2-BL500TNG	M39006/02-1247
CL2-BN010TNE	CL2-BN010TNE	M39006/02-2254
CL2-BN040TNE	CL2-BN040TNG	M39006/02-1248
CL2-BN120TNE	CL2-BN120TNG	M39006/02-1249
CL2-BN250TNE	CL2-BN250TNG	M39006/02-1250
CL2-BN350TNE	CL2-BN350TNG	M39006/02-1251
CL2-BQ0R5SNE	CL2-BQ0R5SNE	M39006/02-2259
CL2-BQ020SNE	CL2-BQ020SNG	M39006/02-1252
CL2-BQ060SNE	CL2-BQ060SNG	M39006/02-1253
CL2-BQ120SNE	CL2-BQ120SNG	M39006/02-1254
CL2-BQ180SNE	CL2-BQ180SNG	M39006/02-1255

1/ Complete type designation will include an additional symbol to indicate style.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See Fig 1.
Case type - Tubular; uninsulated (CL22), insulated (CL23).
Seal type - Hermetic (symbol G).
- Nonhermetic (symbol E).
Terminals - Axial wire-lead.
Rated temperature - -55°C to +85°C, voltage derated to +125°C (See Fig 2).
Rated voltage - See Table I.
Capacitance (cap)(nom) - See Table I.
Capacitance tolerance - Symbol S (+30, -15%), T (+50, -15%), or U (+75, -15%), as shown in Table I.

DC leakage (DCL) (max) (at +25°C):
See Table I.

Dissipation factor (DF)(max):
20% (≤50v) or 15% (>50v).

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - 2,000 vdc.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min): 100 megohms.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25°C value of Table I.
Δ Cap - Within ±5% of initial measurement.
DF (max) - 115% of initial requirement.

Solderability: In accordance with MIL-C-3965.

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Terminal strength:
Full test - In accordance with MIL-C-3965.
Bend test - Not applicable
Torque test - Not applicable
Wire-lead bend test - In accordance with MIL-C-3965.
Twist test - Not applicable.

Surge voltage: See Table I.

Moisture resistance:
DCL (max) - 125% of 25°C value of Table I.
ΔCap - Within ±6% of initial measurement.
DF (max) - 115% of initial requirement.

Low temperature (storage):
DCL (max) - See Table I (25°C).
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25°C):
DCL (max) - See Table I.
Cap- Within tolerance of Table I value.
DF(max) - Shall not exceed initial requirement.
Step 2 (-55°C):
Impedance - See Table I.
ΔCap (max) - -40% (≤50v) or -35% (>50v) of step 1 value.
Step 3 (+25°C):
DCL (max) - See Table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85°C):
DCL (max) - See Table I.
ΔCap (max) - +20% of step 1 value.
DF (max) - 30% (≤50v) or 20% (>50v).

1/ Step 5 (+125°C):
DCL (max) - See Table I.
ΔCap (max) - +45% (≤30v) or +30%(>30v) of step 1 value.
DF (max) - 40% (≤30v) or 30% (>30v).
Step 6 (+25°C):
DCL (max) - See Table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Reverse voltage: Not applicable.

Life at +85°C:
2,000-hr (qualification):
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% of initial measurement.
DF (max) - Shall not exceed initial requirement.
240-hr (group B):
DCL (max) - See Table I (25°C).
ΔCap - Within ±10% of initial measurement.
DF (max) - Shall not exceed initial requirement.
1,760-hr (group C):
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Barometric pressure:
Test voltage - Rated voltage.

1/ Life at +125°C:
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% of initial measurement.
DF (max) - 20% (≤30v) or 15% (>30v).
DWV (min)(CL23 only) - 2000 vdc.
IR (min)(CL23 only) - 100 megohms.

Voltage groups (85°C rated voltage).
I ----------15 to 100 v, incl.
II ----------150v.

1/ 125°C derated voltage referenced.

NOTE: The capacitors covered by this specification sheet are substitutable for those covered by MIL-C-3965/18D. See Table II.

Custodians:
Army - EL
Navy -EC
Air Force - 80

Review Activities:
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Navy - EC, SH
Air Force - 11, 17
DSA - ES

User Activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing Activity:
Air Force - 80

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